

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Ratios, rates, proportional relationships, and units

01

10 answers · Rationale-rich · Teach from doc

Q1

—

Answer not provided in source data — see rationale below.

Choice C is correct. If the butterfly traveled 2,100 miles in 120 days, then it traveled, on average, $\frac{2,100 \text{ miles}}{120 \text{ days}} = 17.5$ miles per day.

Choice A is incorrect. This is approximately the average amount of time, in days, it took the butterfly to fly one mile: $\frac{120 \text{ days}}{2,100 \text{ miles}} = 0.057$ days per mile. Choice B is incorrect and may result from an arithmetic error. Choice D is incorrect. This is the number of hours in a day rather than the number of miles flown per day.

Q2

A

A. 7.5

Choice A is correct. If 2.5 ounces of chocolate are needed for each muffin, then the number of ounces of chocolate needed to make 48 muffins is $48 \times 2.5 = 120$ ounces. Since 1 pound = 16 ounces, the number of pounds that is equivalent to 120 ounces is $\frac{120}{16} = 7.5$ pounds.

Therefore, 7.5 pounds of chocolate are needed to make the 48 muffins.

Choice B is incorrect. If 10 pounds of chocolate were needed to make 48 muffins, then the total number of ounces of chocolate needed would be $10 \times 16 = 160$ ounces. The number of ounces of chocolate per muffin would then be $\frac{160}{48} = 3.33$ ounces per muffin, not 2.5 ounces per

muffin. Choices C and D are also incorrect. Following the same procedures as used to test choice B gives 16.8 ounces per muffin for choice C and 40 ounces per muffin for choice D, not 2.5 ounces per muffin. Therefore, 50.5 and 120 pounds cannot be the number of pounds needed to make 48 signature chocolate muffins.

Q3**423.5**Accept also: $847/2$

The correct answer is 423.5. It's given that $5.5 \text{ yards} = 1 \text{ rod}$. Therefore, 77 rods is equivalent to $77 \text{ rods} \frac{5.5 \text{ yards}}{1 \text{ rod}}$, or 423.5 yards. Note that 423.5 and $847/2$ are examples of ways to enter a correct answer.

Q4**C**

C. 320

Choice C is correct. It's given that the population density of Worthington is 290 people per square mile and Worthington has a population of 92,800 people. Therefore, the area of Worthington is $92,800 \text{ people} \frac{1 \text{ square mile}}{290 \text{ people}}$, which is equivalent to $\frac{92,800 \text{ square miles}}{290}$, or 320 square miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**C****C.** 140

Choice C is correct. The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, then the object's weight on Venus is approximately $\frac{9}{10}(100) = 90$ pounds. The same object's weight on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth; therefore, the object weighs approximately $\frac{23}{10}(100) = 230$ pounds on Jupiter. The difference between the object's weight on Jupiter and the object's weight on Venus is approximately $230 - 90 = 140$ pounds. Therefore, an object that weighs 100 pounds on Earth weighs 140 more pounds on Jupiter than it weighs on Venus.

Choice A is incorrect because it is the weight, in pounds, of the object on Venus. Choice B is incorrect because it is the weight, in pounds, of an object on Earth if it weighs 100 pounds on Venus. Choice D is incorrect because it is the weight, in pounds, of the object on Jupiter.

Q6**C****C.** 5,100

Choice C is correct. Since 1 meter is equal to 100 centimeters, 51 meters is equal to $51 \text{ meters} \frac{100 \text{ centimeters}}{1 \text{ meter}}$, or 5,100 centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from dividing, rather than multiplying, 51 by 100.

Choice D is incorrect. This is the length, in millimeters rather than centimeters, that is equivalent to a length of 51 meters.

Q7**348**

The correct answer is 348. It's given that 1 fathom is equivalent to 6 feet. Therefore, 58 fathoms is equivalent to $58 \text{ fathoms} \frac{6 \text{ feet}}{1 \text{ fathom}}$, or 348 feet. Thus, when the camera is at a depth of 58 fathoms, the camera's depth, in feet, is 348.

Q8**A****A. 5**

Choice A is correct. It's given that the ratio of black pens to red pens is 8 to 1. Therefore, there are $\frac{1}{8}$ as many red pens as black pens in the box. It's also given that there are 40 black pens in the box. Therefore, the number of red pens is $\frac{1}{8}$ of the 40 black pens. Thus, the number of red pens is $40 \frac{1}{8}$, or 5.

Choice B is incorrect. This is the number of black pens in the box for every red pen.

Choice C is incorrect. This is the number of black pens in the box.

Choice D is incorrect. This is the number of red pens in the box if the ratio of black pens to red pens is 1 to 8.

Q9**66**

The correct answer is 66. It's given that a product costs 11.00 dollars per pound. Therefore, the cost for 6 pounds of the product is $\frac{11.00 \text{ dollars}}{1 \text{ pound}} 6 \text{ pounds}$, which is equivalent to 66.00, or 66, dollars.

Q10

A

A. \$551.25

Choice A is correct. At the exchange rate of 1 peso = 0.075 US dollars, 7,500 pesos would be converted to $7,500 \times 0.075 = \$562.50$. However, since Maria pays a 2% fee on the converted US dollar amount, she receives only $(100 - 2)\%$, or 98%, of the converted US dollars, and $562.50 \times 0.98 = \$551.25$.

Choice B is incorrect. This is the number of US dollars Maria would receive if the exchange service did not charge a 2% fee. Choice C is incorrect and may result from a decimal point error made when calculating the conversion to US dollars and from not assessing the 2% fee. Choice D is incorrect and may result from reversing the units of the exchange rate.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q1 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

Generated 2026-05-29 · cb-educator-question-bank